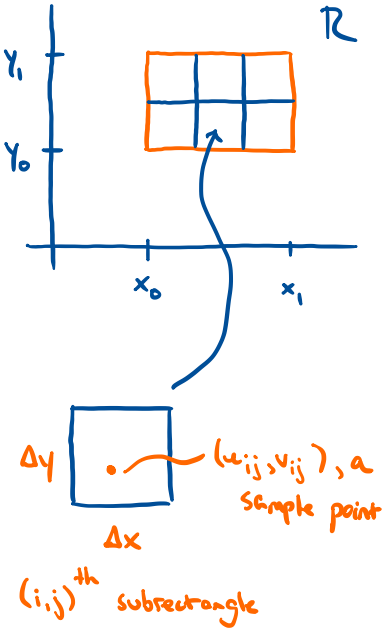
16.1 Integrals of functions of two variables



Suppose R is a rectangular region in the plane and $z = f(x,y)$ is a function. Divide R into smaller rectangles of dimensions Δx by Δy. Then the signed volume of the "cuboid" between the surface of f and the $(i,j)^{th}$ subrectangle is

$$f(u_{ij}, v_{ij})\,\Delta A$$

where $\Delta A = \Delta x \Delta y$ and (u_i, v_j) is a point in the rectangle. Hence the total signed volume between the rectangle R and the surface of f is approximately

$$\sum_{ij} f(u_{ij}, v_{ij})\,\Delta A$$ ← A Riemann sum

This becomes exact in the limit:

← Definition of integral.

$$\int_R f\,\partial A = \lim_{\Delta A \to 0} \sum_{ij} f(u_{ij}, v_{ij})\,\Delta A$$

→ Do all problems.